SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Constituent of Symbiosis International (Deemed University), Pune

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Subject: Web Application and Development	**Sem: III AY: 2026-27**
EXPERIMENT No: 11	**Assignment No: 11**
TITLE: Create a MongoDB Atlas cluster, connect it to a web application, and perform basic CRUD operations	

Required Softwares/Tools:

Software	Purpose
Node.js	Runs your server.js
VS Code	Write/edit server.js and index.html
MongoDB Atlas (online account)	Stores your student records online. It will act as a database.
Web Browser	Opens your application and runs it.
npm (Node Package Manager)	Installs Node.js packages: express, mongodb and dotenv

CONNECTION STRING NEED TO BE CREATED TO CONNECT THE APPLICATION TO MONGODB ATLAS

mongodb+srv://vatsalposty01_db_user:E11_123@cluster0.t4ny2hm.mongodb.net/?appName=Cluster0

we need to add password here. Create a very simple password without special symbols. I will create as E11_123

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open vscode and create a project folder on any of the hard drive on your computer.

Now add server.js and project/index.html files in the newly created folders.
Both files I will share with you.

Initialize nodejs project. Run npm init -y
This will create a basic Node.js project for you
It will generate package.json file in the project folder.

```
Problems  Output  Debug Console  Terminal  Ports

PS D:\SIT\WAD> cd javascript
PS D:\SIT\WAD\javascript> cd E11
PS D:\SIT\WAD\javascript\E11> npm init -y
Wrote to D:\SIT\WAD\javascript\E11\package.json:

{
  "name": "e11",
  "version": "1.0.0",
  "description": "",
  "main": "server.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node server.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}


PS D:\SIT\WAD\javascript\E11>
```

Install Required Packages using node package manager (npm)
Run npm install express mongodb dotenv

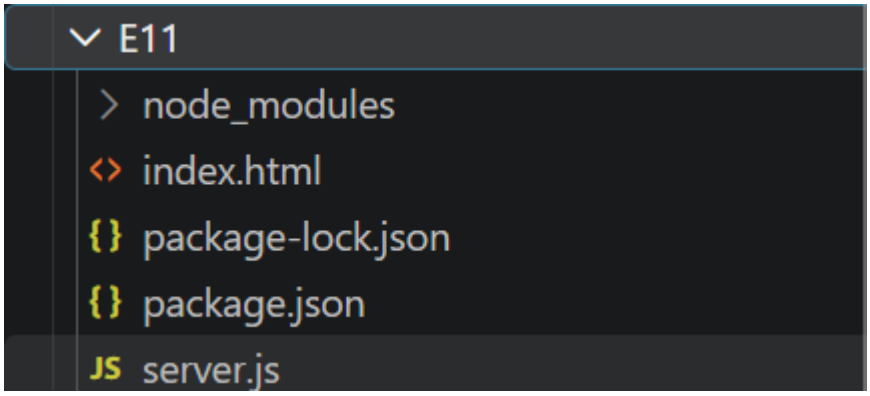
```
PS D:\SIT\WAD\javascript\E11> npm install express mongodb dotenv

added 81 packages, and audited 82 packages in 14s

29 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS D:\SIT\WAD\javascript\E11>
```

After installation, the project will show: node_modules and package-lock.json in the project folder

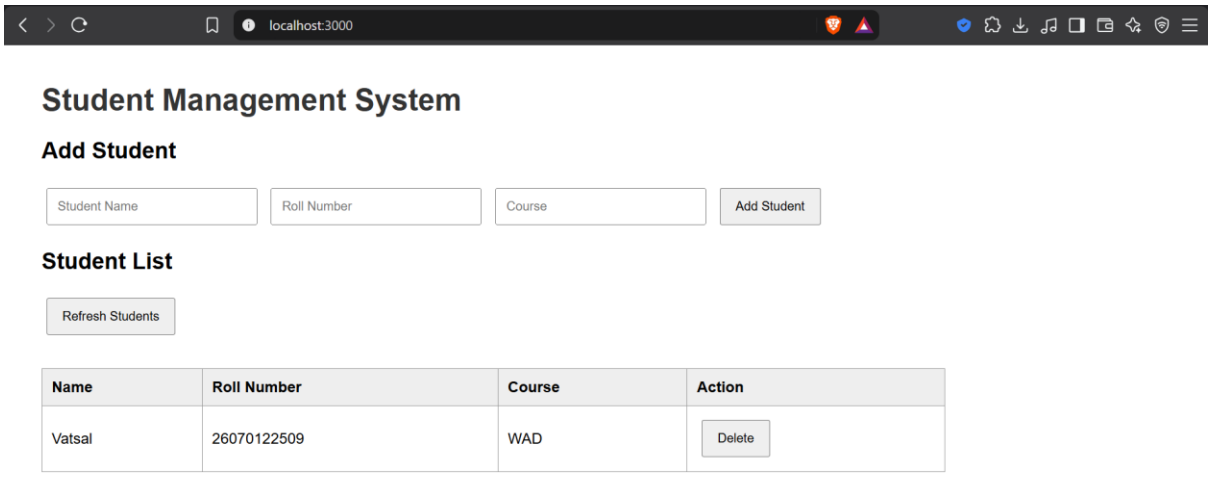


Create .env file in the project folder.
Add connection string into it. We have already created connection string.
For example:
MONGODB_URI=mongodb+srv://vatsalposty01_db_user:E11_123@cluster0.t4ny2hm.mongodb.net/?appName=Cluster0

Do not give any spaces after or before the = sign. No space is allowed in the connection string.

Now, Start the Application by running following command on the terminal:
node server.js

This will start the server.
Now we need to start the browser and need to access the web application's index page.



Student Management System

Add Student

Vatsal | 26070122509 | JAVA | Add Student

Student List

Refresh Students

Name	Roll Number	Course	Action
Vatsal	26070122509	WAD	Delete

Student Management System

Add Student

Student Name | Roll Number | Course | Add Student

Student List

Refresh Students

Name	Roll Number	Course	Action
Vatsal	26070122509	WAD	Delete
Vatsal	26070122509	JAVA	Delete

Student Management System

Add Student

Student Name | Roll Number | Course | Add Student

Student List

Refresh Students

Name	Roll Number	Course	Action
Vatsal	26070122509	WAD	Delete
Vatsal	26070122509	JAVA	Delete
Vatsal	26070122509	FCC	Delete
Vatsal	26070122509	DSA	Delete
Vatsal	26070122509	DBMS	Delete